NEBULA CLOUD STUDIO

# Thermal Assessment — Screening Report

Illustrative solar combiner box • Synthetic demonstration data • Version 0.1

| | |
|---|---|
| Enclosure | **800 × 600 × 300 mm** |
| Mounting | **Wall-mounted** |
| Ambient | **40°C** |
| Assumed air limit | **55°C** |
| Total input losses | **181.4 W** |
| Exposed area | **1.32 m²** |
| Est. internal bulk-air | **67.5°C** |
| Rise above ambient | **27.5 K** |
| Margin versus input limit | **–12.5 K** |
| Preliminary status | **Exceeds illustrative air limit** |

## Engineering observations

- The example uses synthetic device-loss assumptions, not OEM-validated values.
- A simple steady-state heat-balance estimate indicates an air-temperature exceedance.
- Confirm material, installation, outdoor exposure, loss data and component limits.
- Design changes require recalculation and independent engineering review.

**SCOPE & LIMITATIONS**

Preliminary engineering screening only, based on user-entered assumptions.
NOT an IEC TR 60890 calculation, IEC 61439 verification, or compliance certificate.
NOT a physical test, CFD simulation, or product-specific thermal validation.
Do not use these figures for equipment selection or electrical product sign-off.